

Given one image, what can we do

- ① keypoint detection
- ② keypoint refinement if subpixel accuracy is required.
- ③ Gradient Calculation for histogram generation
- ④ 128 D vector descriptor generation

$I_2$

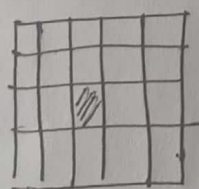
1-4

Same as  $I_1$

Euclidean + Lowe's ratio distance

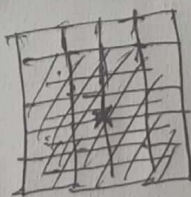
If  $I_{size} < 16 \times 16$

e.g.  $I_{size} = 4 \times 4$ ,  $I_{size} = 8 \times 8$   
of  $8, 32$   
only 1 histogram are generated.



$I$

→ shaded pixel is the one which was declared keypoint  
 $16 \times 16$  window will be. Here  
as shown below where lower right side is reference



$I_{original} \rightarrow$  keypoint  $\rightarrow$  keypoint localized in  $I_{original} \rightarrow$  histogram.

$S_x, S_y \rightarrow \sqrt{S_x^2 + S_y^2}, \tan^{-1}\left(\frac{S_y}{S_x}\right) \rightarrow$

14'

